

MOHAMMED FAIZ NAWAZ

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Hiring Committee
College of Engineering, Robotics Department
University of Michigan
Ann Arbor, MI

Dear Hiring Committee,

I am writing to express my strong interest in the Applications Programmer Associate position within the College of Engineering Robotics Department. I recently completed my Master of Science in Computer Science at the University at Buffalo and bring hands-on experience in software development, web application design, and database-driven automation that align closely with this role.

At Cognizant Technology Solutions, I developed and maintained enterprise-level applications using C Sharp, Python, and JavaScript, improving system performance and streamlining internal workflows. I built full stack dashboards and automation tools that reduced manual debugging time by over 50 hours per quarter and collaborated with multidisciplinary teams to deploy CI CD pipelines for smooth application delivery.

My academic projects further strengthened my ability to design reliable and secure systems. I built a Wikipedia Pageview Analytics pipeline on AWS EMR with Spark and Terraform to process large datasets efficiently. I also implemented an AI-powered Chest X ray classification system that demonstrated how machine learning can enhance accessibility in healthcare automation. These experiences taught me to translate user needs into practical, efficient solutions that improve operations and user experience.

What excites me most about this opportunity is the chance to contribute to a robotics-driven academic environment. Working at the intersection of research, software, and innovation aligns perfectly with my passion for creating technology that supports education and discovery. I am confident that my technical foundation, problem-solving mindset, and collaborative approach would make me a valuable addition to your team.

Thank you for considering my application. I would appreciate the opportunity to discuss how my experience and skills can contribute to the continued success of the Robotics Department.

Kind Regards,

Mohammed Faiz Nawaz
Applicant

Mohammed Faiz Nawaz

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SUMMARY

Graduate Software Engineer with hands-on experience in full-stack development, DevOps, and cloud infrastructure. Proficient in Python, Java, and C++ with expertise in, containerization, and CI/CD pipelines. Delivered scalable data pipelines and AI/ML applications using Spark, PyTorch, and FastAPI, with a track record of optimizing performance, automating workflows, and shipping reliable, production-ready solutions.

SKILLS

Languages: Python, Java, C++, JavaScript, Typescript | **Databases:** SQL Server, Postgres, MongoDB, Redis
Web: Django, React, Flask, HTML, CSS, jQuery, Streamlit | **Cloud:** Azure DevOps, Azure Data Factory, One Lake, AWS EMR, EC2, S3, RDS
Tools: Docker, Git/Gitlab, Jenkins, Kubernetes, Terraform | **Frameworks:** Spark, Hadoop, Postman, selenium Webdriver, Tortoise SVN
Testing: Junit, TestNG, A/B Testing, Integration testing | **AI/ML Frameworks:** PyTorch, scikit-learn, Hugging Face, OpenAI API

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions *(DAM assertSERV by cognizant a PaaS platform that is used to manage digital assets)*

Programmer Analyst Remote, India | Jan 2023 - Jan 2024

- Designed and implemented a real-time tracking system for product releases, seamlessly integrating with **Jenkins** pipelines to provide instant updates and detailed insights into package features, enhancing team collaboration by 80%.
- Developed a **Docker** based **virtualization** container with **Helm** charts and Anaconda support on **Ubuntu**, from scratch within a day.
- Created a full-stack dashboard to manage DevOps **CI/CD** infrastructure, reducing manual troubleshooting by 50 hours per quarter.
- Utilized selenium, Java, and **Page Object Model (POM)** to automate test cases, reducing manual testing efforts by 30%.
- Reduced execution time of each **SQL Query** with refactoring to improve site load time by 5 seconds per page.

Cognizant Technology Solutions (Internship)

Remote, India | Dec 2021 - May 2022

Programmer Analyst Trainee

- Developed, enhanced, tested a **.NET** web application using **C#** and **EntityFramework**, improving application performance and reliability.
- Completed **POCs** to improve the performance of multiple data-intensive systems, enhancing legacy web page load times by 40%.
- Collaborated with a team of 15 people and enriched existing system metrics by 17%.
- Built a website using **.NET CORE** with an **ORACLE database**, enabling the processing of large volumes of records efficiently and facilitating smooth user operations.
- Conducted POCs to test multiple strategies for migrating data from **Oracle DB** to **Snowflake DB**, achieving minimal cost and improved efficiency.

PROJECTS

Big Data Analytics – Wikipedia (Python, Spark, AWS EMR, S3)

- Performed **ETL** on a 128GB Wikipedia dataset using **Python** and **Bash**, preparing it for distributed processing and analysis.
- Developed and deployed a **Spark** Streaming application on **AWS EMR** to process and aggregate real-time page view data.
- Engineered **EMR** cluster provisioning with **Terraform**, enabling scalable, cost-efficient, and reproducible data workflows.
- Conducted trend analysis using **pandas** in Jupyter Notebook, extracting daily search insights and identifying event-driven spikes.

Loan Default Prediction – (Python, numpy, pandas, Streamlit)

- Developed a machine learning model using historical data of 10,000+ records to predict loan defaults, implementing algorithms like Logistic regression, Decision Tree, KNN, Naive Bayes, and Random Forest.
- Optimized the model by reducing features from 54 to 10 using Recursive Feature Elimination and Select K Best, achieving 90% accuracy, 0.88 F1-score, and a 20% reduction in error rates through hyperparameter tuning.

FitFuel (AI Diet App) – (GPT-2, FastAPI, Docker)

- AI-powered meal planner using **DistilGPT-2** trained on nutrition data, served via **FastAPI** with a **Next.js UI**.
- Dockerized full-stack app deployed on **DigitalOcean** with **NGINX** reverse proxy for scalability and routing.
- Implemented prompt engineering and input validation for accurate, user-specific diet chart generation.

Chest X-ray Prediction – (CNN, FastAPI, Next.js)

- Medical image classifier (94% accuracy) using **CNN** on **ChestMNIST**, served via **FastAPI** with a Next.js frontend.
- Fully Dockerized application deployed for reliable, scalable, and public inference.
- Applied image preprocessing and data augmentation techniques to enhance model generalization and robustness

EDUCATION

SUNY - University at Buffalo

Buffalo, United States

Master of Science (MS) in Computer Science – 3.67 GPA

Jan 2024 – June 2025

- Relevant Coursework:** Database systems, Modern Concepts Networking, Operating Systems, Machine learning, Computer Security, Data Intensive Computing, Project Management

Jawaharlal Nehru Technological University, Kakinada

Kakinada, India

Bachelor of Technology in Electrical and Electronics Engineering – 3.5 GPA

Aug 2018 – May 2022

- Relevant Coursework:** Computer Programming, Data structures, Algorithms, Data Mining, Microprocessors, Cryptography, Digital Design

LEADERSHIP SKILLS

Student Ambassador – UX

Part time | Buffalo, NY, United States

- Participated in User research by collecting data on User inquiries, preferences, pain points, use of space, and more.
- Assisted the User Experience and Information Desk Services team with projects as needed.
- Participated in UB outreach events such as tabling at Accepted Students Days, filming promotional videos.